

## Database Design and Table Relations

The database uses three tables: users, expeditions, and bookings. The schema is intentionally small, but it captures both operational data and reservation transactions.

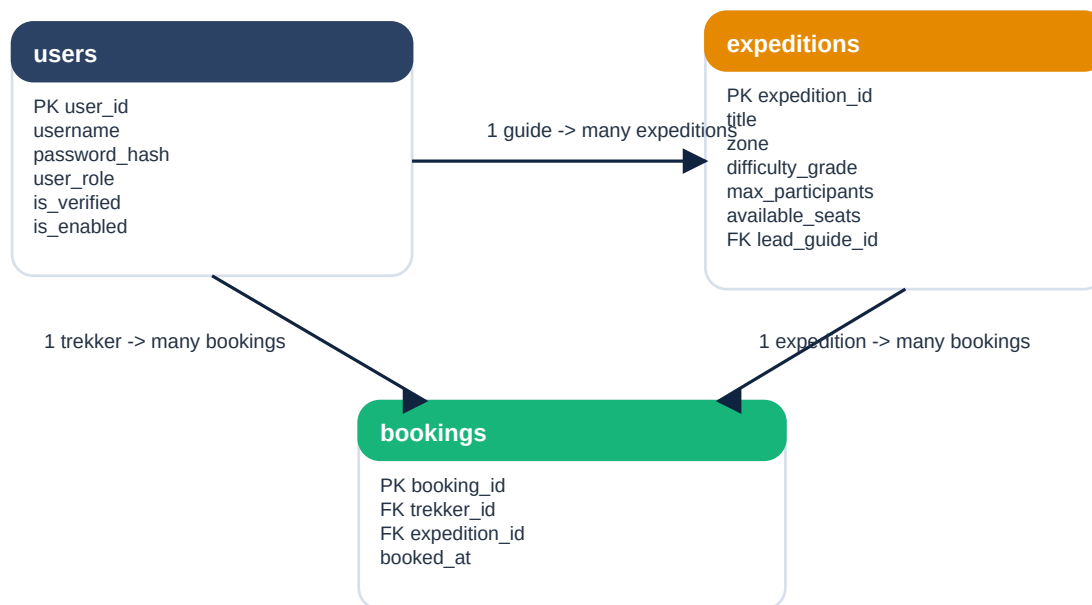


Figure 2. ER diagram with table relationships.

### Relations Between Tables

- users -> expeditions: one guide user can be assigned to many expeditions through `expeditions.lead_guide_id`. This is a one-to-many relationship.
- users -> bookings: one trekker user can create many bookings through `bookings.trekker_id`. This is a one-to-many relationship.
- expeditions -> bookings: one expedition can appear in many booking records through `bookings.expedition_id`. This is a one-to-many relationship.
- users and expeditions are therefore connected indirectly for trekkers through the bookings bridge table, creating a many-to-many participation pattern.

### Data Dictionary

| Table       | Important Fields                                                                                | Purpose                                                                               |
|-------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| users       | user_id, username, password_hash, user_role, is_verified, is_enabled                            | Stores authentication and account lifecycle state for managers, guides, and trekkers. |
| expeditions | expedition_id, title, zone, difficulty_grade, available_seats, expedition_status, lead_guide_id | Stores expedition catalog information and operational ownership.                      |
| bookings    | booking_id, trekker_id, expedition_id, booked_at                                                | Stores reservation transactions between trekkers and expeditions.                     |